



X-ray linear dichroism study on $S = 2$ antiferromagnets $\text{Ba}_2\text{FeSi}_2\text{O}_7$ and $\text{Sr}_2\text{FeSi}_2\text{O}_7$: determination of single-ion magnetic anisotropy energies

Jong-Hyeok Choi^{1,2} · Dae-Hyeon Bahng^{1,3} · Tae-Hwan Jang² · Jae-Hoon Park^{1,2}

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Abstract

We investigated $S = 2$ quasi-two-dimensional XY Heisenberg antiferromagnets $\text{Ba}_2\text{FeSi}_2\text{O}_7$ (BFSO) and $\text{Sr}_2\text{FeSi}_2\text{O}_7$ (SFSO) using the Fe $L_{2,3}$ -edge X-ray linear dichroism (XLD). The XLD spectrums exhibit nearly identical spectral line shapes with small variations for BFSO and SFSO, which form isostructural square-lattice magnetic networks of compressed FeO_4 (d^6) tetrahedrons. For the XLD line shape analysis, we performed the full multiplet cluster model calculations and successfully reproduced not only the spectral line shape but also small variations through different degrees of the compressive distortion for BFSO and SFSO. The model calculations confirm that the $S = 2$ 5E_g ($t_{2g}^3 e_g^3$) state is split into $S_z = 0, \pm 1$, and ± 2 by the spin–orbit coupling. In the analysis, we found that the splitting energy D , denoted as the single-ion magnetic anisotropy energy of the order of 1 meV scale, varies with the degree of the distortion and determined $D_{\text{BFSO}} = 1.45$ meV and $D_{\text{SFSO}} = 1.22$ meV.

Keywords X-ray absorption spectroscopy · Cluster model calculation · $\text{Ba}_2\text{FeSi}_2\text{O}_7$

1 Introduction

X-ray absorption spectroscopy (XAS) is a powerful, element-specific probe of localized electronic states. In transition metal compounds, XAS directly provides information on the valence state, and when combined with polarized light, it becomes sensitive to the spin and orbital degrees of freedom. X-ray magnetic circular dichroism (XMCD) allows one to determine spin and orbital magnetic moments, whereas X-ray linear dichroism (XLD) reveals anisotropic orbital occupation [1–7]. By comparing XMCD and XLD spectra with full multiplet cluster calculations, one can quantify splitting energies induced by physical environments such

as local distortions, spin–orbit coupling, etc., in spite of their energy scales being much smaller than the experimental energy resolution.

Recently, a quasi-two-dimensional XY Heisenberg antiferromagnet $\text{Ba}_2\text{FeSi}_2\text{O}_7$ (BFSO) and its isostructural material $\text{Sr}_2\text{FeSi}_2\text{O}_7$ (SFSO) have been identified as a quantum magnetism model system in which the Fe^{2+} (d^6) ion realizes an $S = 2$ manifold split into $S_z = 0, \pm 1$, and ± 2 levels [8–11]. BFSO crystallizes in the noncentrosymmetric tetragonal $\text{P}\bar{4}2_1m$ space group symmetry with lattice parameters $a = b = 8.3261$ Å and $c = 5.3402$ Å [12]. The Fe^{2+} ions reside in compressed FeO_4 tetrahedra that form magnetic layers separated by Ba slabs together with nonmagnetic SiO_4 units as illustrated in Fig. 1a. The FeO_4 tetrahedra are compressed along the crystallographic c axis, and the Fe–O–c angle deviates from the ideal tetrahedral value by 7.84° in BFSO and 4.86° in SFSO due to the rather large A-site (Ba and Sr) ionic size. This layered arrangement and the dominant Fe–O–O–Fe superexchange pathway render the magnetism highly anisotropic and quasi two-dimensional. Antiferromagnetic (AFM) ground state is stabilized below $T_N = 5.2$ K (BFSO) or below $T_N = 5.0$ K (SFSO) with a collinear propagation vector $Q = (1, 0, 1/2)$ [12, 13].

✉ Jae-Hoon Park
jhp@postech.ac.kr

¹ Department of Physics, POSTECH, Pohang 37673, Republic of Korea

² Center for Complex Phase Materials, Max Planck POSTECH/Korea Research Initiative, Pohang 37673, Republic of Korea

³ Institute for Basic Science (IBS), Daejeon 34141, Republic of Korea

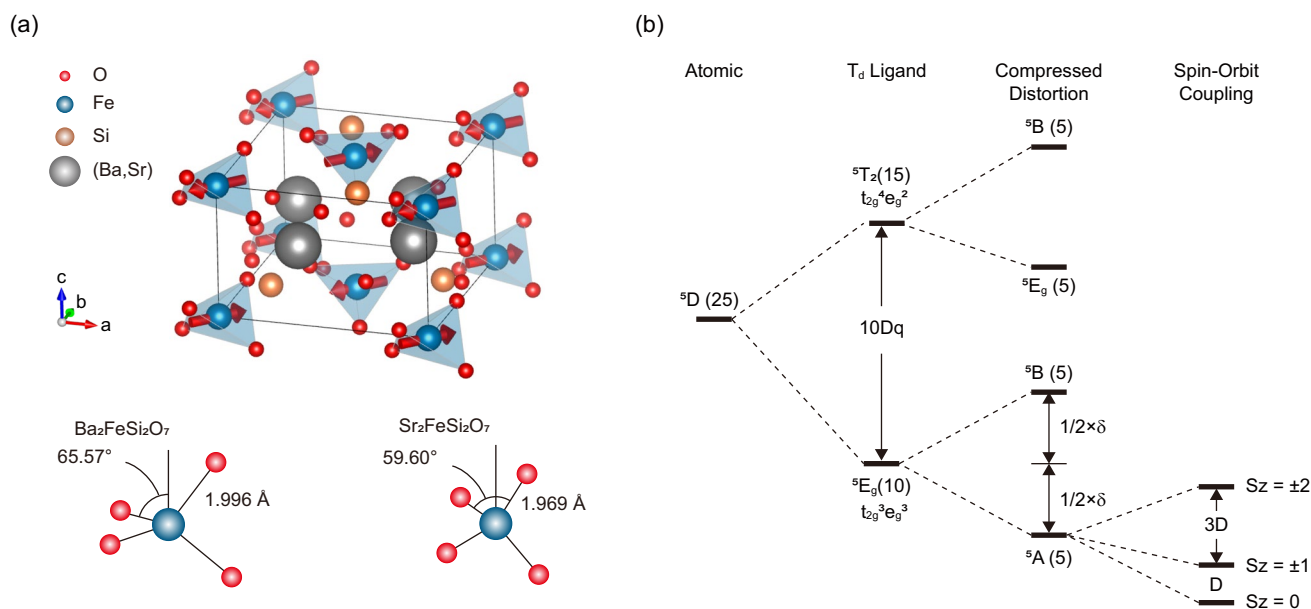


Fig. 1 **a** Magnetic structure of $\text{Ba}_2\text{FeSi}_2\text{O}_7$ below $T_N = 5.2$ K [11]. The FeO_4 figure indicates the degrees of Fe–O distance angle of $\text{Ba}_2\text{FeSi}_2\text{O}_7$ and $\text{Sr}_2\text{FeSi}_2\text{O}_7$ tetrahedron between O–Fe–O resulting

from compressed distortion. **b** The energy diagram for a compressively distorted tetrahedron of Fe^{2+} (d^6) ion

Figure 1b depicts the schematic energy diagrams of the ionic energy level splitting of the $S = 2$ Fe^{2+} ($3d^6$) state, which evolve with the tetrahedral T_d ligand field, additional compressive distortion, and the Fe 3d spin–orbit coupling (SOC), reflecting the compressed FeO_4 tetrahedron in BFSO and SFSO. As can be seen in the figure, the lowest energy ionic state becomes the 5E_g ($e_g^2 t_{2g}^3 e_g^3$) state with an additional minority spin $e_{g\downarrow}$ electron to the half-filled majority spin electron configuration 6A_1 ($e_g^2 t_{2g}^3$) state. The compressive tetragonal distortion of tetrahedron splits the degenerated $d_{3z^2-r^2}$ and $d_{x^2-y^2}$ e_g orbital states, and the lowest energy state becomes the 5A state with a $d_{3z^2-r^2}$ electron. As the Fe 3d spin–orbit is turned on, the fivefold 5A state couples with the high energy 5E_g ($e_g^2 t_{2g}^3 t_{2g\downarrow}$) state with either a $d_{yz\downarrow}$ or a $d_{zx\downarrow}$ electron and splits again into the $S_z = 0, \pm 1$, and ± 2 states described by the so called, single-ion magnetic anisotropy DS_z^2 [14], resulting in the $S_z = 0$ quantum paramagnetic (QPM) ground state. In a real solid, the single-ion anisotropy energy D is modified by the Fe 3d–O 2p hybridization, and the Heisenberg exchange energy J from the magnetic network lowers the $S_z = \pm 1$ AFM state [10, 11]. As the result of these two competing terms D and J both with the order of 1 meV scale, the ground state is expected to experience a phase transition between the $S_z = \pm 1$ AFM and the $S_z = 0$ QPM state across a critical ratio $D_C/J \equiv \alpha_C$. Hence, the determination of the D value is very crucial to understand the ground state.

In this paper, we elucidate the Fe 3d multiplet structures of BFSO and SFSO and evaluate the meV-scale D value by

combining Fe $L_{2,3}$ -edge XAS and XLD on single crystals with distortion-dependent full multiplet cluster calculations. We further investigate the D values with variation in the degree of the tetragonal distortion and predict the critical degree of the distortion corresponding to $D = D_C$, which can be approached through strain control [15].

2 Methods

2.1 Full multiplet cluster model calculation

We modeled the FeO_4 units using full multiplet cluster calculations implemented in the XTLS code [16]. The intra-atomic Coulomb and exchange integrals interaction ($F_{dd}^2, F_{dd}^4, F_{pd}^1, F_{pd}^3, G_{pd}^1, G_{pd}^3$) were reduced to 82.5% of their Hartree–Fock values [17]. Since BFSO and SFSO are isostructural, we used the same on-site Coulomb interactions and charge-transfer energy for both compounds, setting $U_{dd} = 4.5$ eV, $U_{pd} = 5.75$ eV, and $\Delta = 5.5$ eV [18]. The crystallographic Fe–O bond lengths $R_{\text{BFSO}} = 1.996$ Å and $R_{\text{SFSO}} = 1.969$ Å determined for BFSO and SFSO were taken into account in the model calculations by adjusting the p – d hybridization strengths with the Harrison’s rule [12, 13, 19–23]. The single-ion anisotropy energies D were calculated as a function of the degree of the compressive tetragonal distortion in the FeO_4 tetrahedron represented by the deviation angle $\Delta\theta$ in a tetrahedral angle θ between the crystallographic c axis and the Fe–O bonds from the ideal

$\theta_0 = 54.74^\circ$ in a perfect tetrahedron under a fixed Poisson ratio of $\Delta\theta$ to the increase of the Fe–O bond length determined from the BFSO and SFSO cases.

2.2 X-ray absorption spectroscopy and X-ray linear dichroism

Fe $L_{2,3}$ -edge XAS and XLD spectra of single-crystalline BFSO and SFSO were measured at the 2A beamline of the Pohang Accelerator Laboratory and normalized by the edge jump. The experimental geometry was defined such that the crystallographic c axis was aligned parallel to the incident photon wave vector $\mathbf{k}$. The samples were cleaved in situ under a base pressure of 8×10^{-9} Torr, and all measurements were carried out at $T = 370$ K under identical ultrahigh-vacuum conditions with total electron yield mode. For the XLD measurements, the crystals were rotated by 67.5° within the ab – c scattering plane. The energy resolution at the Fe $L_{2,3}$ -edges was approximately 0.1–0.2 eV.

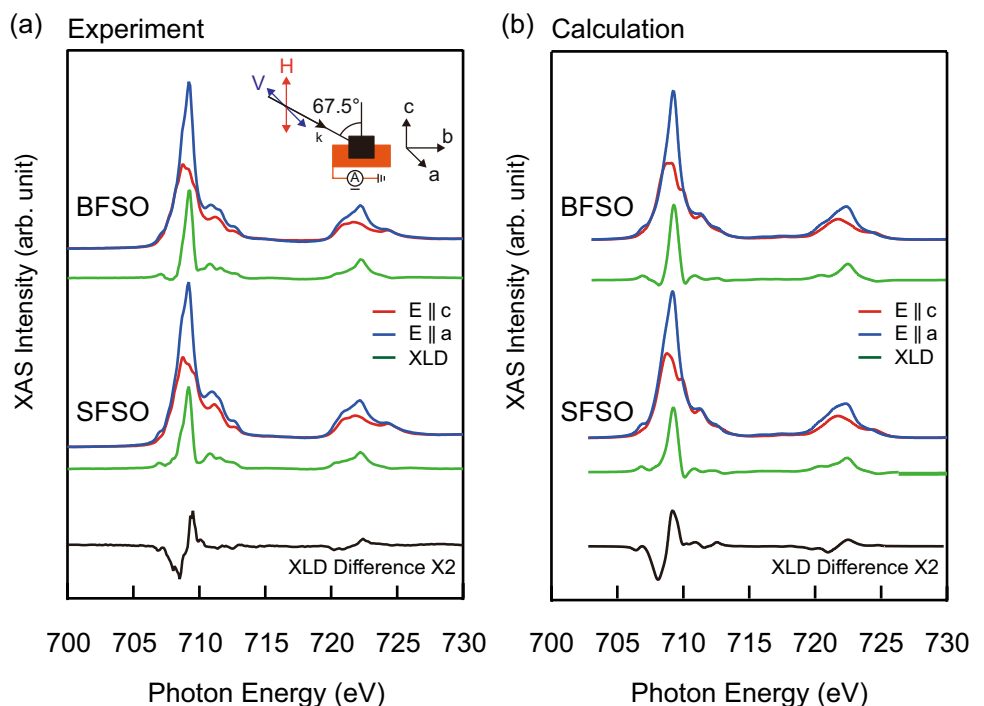
3 Results and discussion

Figure 2a shows the polarization-dependent Fe $L_{2,3}$ -edge XAS and XLD spectra of BFSO and SFSO measured with the incident light of different electric field $\mathbf{E}$ polarization vectors parallel to the crystal c axis ($\mathbf{E} \parallel c$) and to the a axis ($\mathbf{E} \parallel a$). Both BFSO and SFSO exhibit strong XLD signals $I(\mathbf{E} \parallel a) - I(\mathbf{E} \parallel c)$ with very similar line shape, reflecting the orbital anisotropy in the layered quasi-2D structural

character with a very similar local FeO_4 compressive tetrahedron environment. Considering that the XAS spectrum reflects the unoccupied Fe $3d$ states, the positive XLD signal represents the unoccupied orbital character predominated by the in-plane orbitals, in consistency with the expected 5A ionic ground state with the $d_{3z^2-r^2}$ occupation. Meanwhile, one can also recognize small variations in the XLD signal in the two systems, and the difference spectrum ($\text{XLD}_{\text{BFSO}} - \text{XLD}_{\text{SFSO}}$), which reflects the variation of the orbital characteristics due to the difference in the degree of the compressive distortion appearing in BFSO and SFSO crystal structures.

To elucidate quantitatively how the compressed FeO_4 tetrahedron contributes to the orbital characteristics and energetics of the ground state, we performed the full multiplet cluster model calculation analysis on the XAS and XLD spectra. Figure 2b shows the Fe $L_{2,3}$ -edge X-ray absorption spectra calculated for BFSO and SFSO cases. In the model calculations, we explicitly take into account the local environments of the FeO_4 tetrahedrons using the crystallographic values of the tetragonal distortion angles and the Fe–O bond lengths, which are $\Delta\theta = 7.84^\circ$ and $R_{\text{BFSO}} = 1.996$ Å in BFSO and $\Delta\theta = 4.86^\circ$ and $R_{\text{SFSO}} = 1.969$ Å in SFSO [12, 13]. As can be seen in the figure, the model calculations well reproduce not only the polarization-dependent XAS and XLD spectra of BFSO and SFSO but also the XLD difference spectrum $\text{XLD}_{\text{BFSO}} - \text{XLD}_{\text{SFSO}}$, demonstrating that the model calculations provide consistent descriptions for the electronic structures including the orbital characteristics of BFSO and SFSO.

Fig. 2 **a** The polarization-dependent Fe $L_{2,3}$ -edge XAS spectra with $\mathbf{E} \parallel c$ (Red) and $\mathbf{E} \parallel a$ (Blue) and the XLD spectrum $I(\mathbf{E} \parallel a) - I(\mathbf{E} \parallel c)$ (Green) for $\text{Ba}_2\text{FeSi}_2\text{O}_7$ and $\text{Sr}_2\text{FeSi}_2\text{O}_7$ and **b** the corresponding model calculation results. The black lines show the difference ($\text{XLD}_{\text{BFSO}} - \text{XLD}_{\text{SFSO}}$) in the XLD spectra of $\text{Ba}_2\text{FeSi}_2\text{O}_7$ and $\text{Sr}_2\text{FeSi}_2\text{O}_7$



The orbital occupation of the additional $e_g \downarrow$ mostly contributes to the orbital anisotropy, and we obtained its occupation to be $d_{3z^2-r^2}^2 = 0.989$ and $d_{x^2-y^2}^2 = 0.030$ for BFSO and to be $d_{3z^2-r^2}^2 = 0.995$ and $d_{x^2-y^2}^2 = 0.032$ for SFSO. In both systems, the $d_{3z^2-r^2}^2$ out-of-plane orbital occupation is dominant to that of the in-plane $d_{x^2-y^2}^2$ orbital, consistent with the polarization-dependent XAS, and the orbital anisotropy is barely reduced from the occupations $d_{3z^2-r^2}^2 = 1$ and $d_{x^2-y^2}^2 = 0$ expected in the ionic case without SOC. Besides the orbital anisotropy, the model calculations also provide quantitative information on the energetics of the diagram in Fig. 1b and enable us to determine the single-ion anisotropy energies $D_{\text{BFSO}} = 1.45$ meV and $D_{\text{SFSO}} = 1.22$ meV, in a good agreement with the previously reported value $D_{\text{BFSO}} = 1.42$ meV determined from the inelastic neutron scattering [10, 11]. The anisotropy energy D originates from the spin-orbit coupling between the 5A state and the 5E_g state (see Fig. 1b), i.e., $D \sim \lambda^2/\Delta E$ with the SOC constant λ and the 5A – 5E_g energy separation ΔE . As the compressive tetragonal distortion increases, ΔE decreases to enhance D . Hence, the smaller $\Delta\theta$ in SFSO increases ΔE by about 20% from that in BFSO, resulting in about a 20% smaller D value.

The neutron scattering study reported the exchange energy $J \approx 0.26$ meV and $D_{\text{BFSO}} = 1.42$ meV, which is somewhat smaller than the critical value $D_c = 1.65$ meV corresponding to α_c proposed theoretically [10, 11]. Finally, we expand the model calculations to predict the D value as a function of the compressive tetragonal distortion angle $\Delta\theta$ with even though its energy scale of the order of 1 meV is two orders of magnitude smaller than the XAS energy resolution of a few tenth of eV. For the calculations, we first introduce a Poisson ratio ν for the local tetragonal distortion of tetrahedron, which is defined by the ratio of the out-of-plane (z -axis) compression ($z_0 - \Delta z$) to the in-plane (x or y axis) expansion ($x_0 + \Delta x$), i.e., $\nu \equiv d(\Delta x)/d(\Delta z)$, yielding $\nu \cong 0.5$ for a fixed volume of the tetrahedron, while this ν -value turns out to be $\nu = 0.492$ in both BFSO and SFSO reduced only by 1.6%, implying that the chemical pressure induced by the A-site (Ba or Sr) substitution acts as like a near-uniaxial pressure. Figure 3 shows the calculation results of the single-ion anisotropy energy D as a function of the distortion angle $\Delta\theta$ in a range of $4^\circ < \Delta\theta < 12^\circ$ under an assumption of a fixed ν . As $\Delta\theta$ increases, the D value monotonically increases to approach the critical value $D_c = 1.65$ meV at $\Delta\theta \approx 10^\circ$ for a fixed $J \approx 0.26$ meV reported for BFSO [10, 11]. With the uniaxial type tetragonal compression, the Fe–O bond length increase to reduce the hybridization strength while the Fe–O–O–Fe hopping angle also increases to enhance the inter-site hopping matrix element. The Heisenberg exchange J decreases by the former effect but increases by the latter effect. Hence, the variation in the J value with the distortion can be minimal as expected

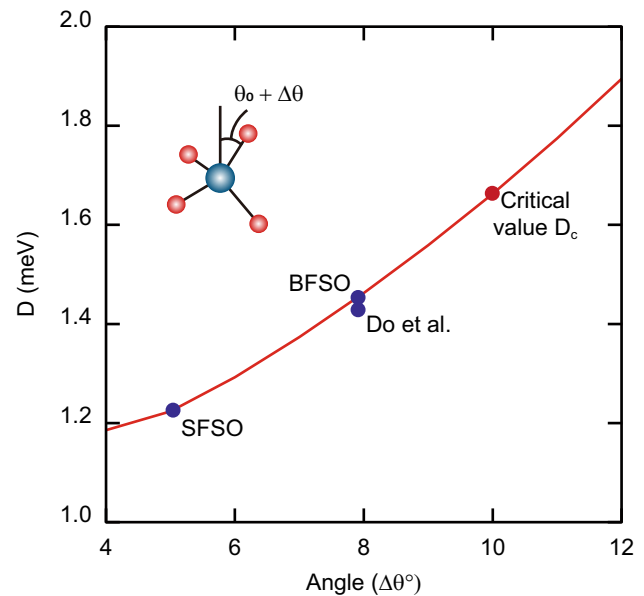


Fig. 3 Single-ion magnetic anisotropy energy D as a function of the deviation angle $\Delta\theta$ from the ideal $\theta_0 = 54.74^\circ$ in a perfect tetrahedron obtained using the cluster model calculations with a fixed Poisson ratio $\nu = 0.492$ as described in the text. The D value is quoted from the previous studies by Do et al. for BFSO [10, 11]. The critical value $D_c = 1.65$ meV can be achieved at $\Delta\theta \approx 10^\circ$ in the model calculations

from the observed similar T_N 's of BFSO ($T_N = 5.2$ K) and SFSO ($T_N = 5.0$ K) [12, 13].

4 Conclusion

We performed the Fe $L_{2,3}$ -edge XAS and XLD studies on $S = 2$ quasi-2D AFM $\text{Ba}_2\text{FeSi}_2\text{O}_7$ (BFSO) and $\text{Sr}_2\text{FeSi}_2\text{O}_7$ (SFSO). The spectra were analyzed using the full multiplet cluster model calculations, in which the crystallographic local environments of the FeO_4 tetrahedra are explicitly taken into account. The distortion-dependent model calculation analysis enables us to determine even the meV-scale single-ion magnetic anisotropy energies of $D = 1.45$ meV for BFSO with the distortion angle $\Delta\theta = 7.84^\circ$ and $D = 1.22$ meV for SFSO with $\Delta\theta = 4.86^\circ$. We also found in the model calculations that D approaches the critical value $D_c \approx 1.65$ meV across the quantum paramagnetic (QPM) and antiferromagnetic (AFM) ground states at a larger $\Delta\theta \approx 10^\circ$.

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Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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